When to use API Keys

Managing access and safeguarding resources is where API keys come into play. An API may require API keys for part or all of its methods. In this reading, we delve into the concept of API keys, exploring their pivotal role in not only securing and controlling access, but also in gathering insights to some processes API keys should not be used for.

There are a few reasons why you might want to use API keys.

What you can use API keys for

Some of the ways you might use API keys include:

- **To block anonymous traffic** - Can help to protect your API from abuse and to ensure that only authorized users are able to access it.
- **To control the number of calls made to your API** - Can help to prevent your API from being overloaded and to ensure that it is available to all authorized users.
- **To identify usage patterns** - Can be used to improve your API and to make sure that it is meeting the needs of your users.
- **To filter logs by API key** - Can help you to troubleshoot problems with your API and to identify which users are using your API the most.

What you cannot use API keys for

You can’t use API keys for:

- **Identifying individual users** - API keys do not identify individual users; they identify entire projects.
- **Secure authorization** - They should be used only to identify and control access to an API.

- **Identifying the creators of a project** - Service Infrastructure doesn't provide a method to directly look up projects from API keys.

Key Takeaways

- You use API keys for blocking anonymous traffic, controlling the number of calls made to your API, identifying usage patterns, and to filter logs by API keys.
- You can’t use API keys for identifying individual users, securing authorization, and identifying the creators of a project.

  API keys serve as the link between the potential of APIs and the demand for restricted usage. As developers continue to harness the power of APIs to weave intricate software ecosystems, a nuanced understanding of API keys' capabilities and boundaries becomes the cornerstone of ensuring secure, efficient, and insightful API management.